

Assignment 1

Submission Link: <https://forms.gle/dVZ1qKdyE2RticxU7>

1. Perform the following base conversions
 - (a) $(10110010101)_{2s} = (?)_7$
 - (b) $(00110010101)_{1s} = (?)_9$
 - (c) $(0011)_{BCD} = (?)_5$
 - (d) $(010010101011111)_{2s} = (?)_{10}$
 - (e) $(101010101000011)_{2s} = (?)_{10}$
 - (f) $(010010101011111)_{1s} = (?)_{10}$
 - (g) $(101010101000011)_{1s} = (?)_{10}$
2. Suppose you have two 4-bit registers where your computer stores signed numbers. Now subtract 5 from -6 using your computer's registers, and figure out whether there is any overflow or not.

Note: The computer uses 2's complement representation for signed numbers.

3. What is the largest number you can store in a 12 bit register using a 1s complement system. Your final answer should be in octal.
4. What is the smallest number you can store in a 8 bit register using a sign & magnitude system. Your final answer should be in HEX.
5. Divide 43434212_5 by 124_5 ; Find the quotient and remainder if any.